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ART UNIT	PAPER NUMBER
1677	

NOTIFICATION DATE	DELIVERY MODE
10/29/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/283,274	Applicant(s) Kobayashi et al.	
	Examiner ELIZABETH OGUNTADE	Art Unit 1677	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 21 August 2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,4,6 and 8 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,4,6 and 8 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 21 September 2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 10/07/2026.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of the Claims

Claims 1-8 are pending. Claims 1, 4, and 6 are amended. Claims 2, 3, 5, and 7 are cancelled. Claims 1, 4, 6, and 8 are examined herein.

Priority

The present application, filed 09/21/2023, is a 371 of PCT/JP2022/013234, filed 03/22/2022, which claims foreign priority of JP2021-049931, filed 03/24/2021. Should applicant desire to obtain the benefit of foreign priority under 35 U.S.C. 119(a)-(d) prior to declaration of an interference, a certified English translation of the foreign application must be submitted in reply to this action. 37 CFR 41.154(b) and 41.202(e).

Failure to provide a certified translation may result in no benefit being accorded for the non-English application.

Information Disclosure Statement

The Information Disclosure Statement(s) filed 10/07/2026 are acknowledged and have been considered.

Claim Rejections - 35 USC § 112

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or

> with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

The previous rejection under 35 U.S.C. §112(a) for failure to comply with the enablement requirement is withdrawn in view of Applicant's amendments to the claims. A new rejection under 35 U.S.C. §112(a) for failure to comply with the written description requirement is made to address the scope of the antibody CDR combinations recited in amended claim 1.

Claims **1**, **4**, **6**, and **8** are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

Regarding claim interpretation, under the broadest reasonable interpretation, claim 1 is directed to a method for determining malignant pancreatic cystic tumor in a test subject having pancreatic cystic tumor comprising the use of monoclonal antibodies to APOA2-AT terminus, APOA2-ATQ terminus, and APOA2 non-terminus, each defined by independently recited alternative heavy-chain CDR1, CDR2, and CDR3 sequences and alternative light-chain CDR1, CDR2, and CDR3 sequences. The term **respectively** associates each sequence identifier with CDR1, CDR2, or CDR3 within the respective heavy or light chain but does not expressly require the heavy-chain CDR triplet of one disclosed antibody clone to be paired exclusively with the light-chain CDR triplet of that same clone. Accordingly, claim 1 is interpreted as encompassing antibodies comprising independently selected heavy-chain and light-chain CDR triplets,

including combinations different from those of the specifically disclosed antibody clones, while requiring the resulting antibodies to possess the recited APOA2 binding specificities.

Regarding the anti-APOA2-AT terminus monoclonal antibody, the specification discloses that specific examples of such an antibody include anti-APOA2-AT terminus monoclonal antibody clones represented by antibody clone names 4C6-1, 5D9-3, and 6B4-2 ([0032], p.18). Paragraph [0032] identifies the corresponding heavy-chain and light-chain CDR sequences of these three clones. Specifically, clone 4C6-1 comprises heavy-chain CDRs represented by SEQ ID NOs. 33–35 and light-chain CDRs represented by SEQ ID NOs. 36–38; clone 5D9-3 comprises heavy-chain CDRs represented by SEQ ID NOs. 39–41 and light-chain CDRs represented by SEQ ID NOs. 42–44; and clone 6B4-2 comprises heavy-chain CDRs represented by SEQ ID NOs. 45–47 and light-chain CDRs represented by SEQ ID NOs. 48–50.

Paragraph [0032] further discloses that the antibody specifically binds to the C-terminal region of the APOA2-AT protein shown in SEQ ID NO: 1, but exhibits no binding activity against the APOA2-ATQ protein shown in SEQ ID NO: 2 and the APOA2-A protein shown in SEQ ID NO: 3. However, claim 1 encompasses additional combinations, including an antibody comprising heavy-chain CDRs represented by SEQ ID NOs. 33–35 and light-chain CDRs represented by SEQ ID NOs. 42–44. This combination differs from the specifically disclosed antibodies. Although the specification identifies the constituent CDR sequences, paragraph [0032] does not establish possession of this additional combination as an antibody possessing the required APOA2-AT binding specificity.

Regarding the anti-APOA2-ATQ terminus monoclonal antibody, the specification discloses that specific examples of such an antibody include anti-APOA2-ATQ terminus monoclonal antibody clones represented by antibody clone names 7F2 and 6G2 ([0031], p.17). Paragraph [0031] identifies clone 7F2 as comprising heavy-chain CDRs represented by SEQ ID NOs. 4–6 and light-chain CDRs represented by SEQ ID NOs. 7–9, and clone 6G2 as comprising heavy-chain CDRs represented by SEQ ID NOs. 10–12 and light-chain CDRs represented by SEQ ID NOs. 13–15.

Paragraph [0031] further discloses that the antibody specifically binds to the C-terminal region of the APOA2-ATQ protein shown in SEQ ID NO: 2, but exhibits no binding activity against the APOA2-AT protein shown in SEQ ID NO: 1 and the APOA2-A protein shown in SEQ ID NO: 3. However, claim 1 additionally encompasses antibodies comprising heavy-chain CDRs represented by SEQ ID NOs. 4–6 and light-chain CDRs represented by SEQ ID NOs. 13–15. The specification identifies the individual sequences and corresponding disclosed clones but does not establish possession of this additional cross-paired combination as an antibody possessing the required APOA2-ATQ binding specificity.

Regarding the anti-APOA2 non-terminus monoclonal antibody, the specification discloses that specific examples thereof include anti-APOA2 antibody clones represented by antibody clone names MAB1 and MAB2 ([0033], p.18). Paragraph [0033] identifies clone MAB1 as comprising heavy-chain CDRs represented by SEQ ID NOs. 16–18 and light-chain CDRs represented by SEQ ID NOs. 19–21, and clone MAB2 as comprising heavy-chain CDRs represented by SEQ ID NOs. 22–24 and light-chain CDRs represented by SEQ ID NOs. 25–27. Paragraph [0033] further discloses that the anti-APOA2-ATQ non-terminus antibody or the anti-APOA2-AT non-terminus antibody can be used as the anti-APOA2 protein non-terminus antibody. However, claim 1 encompasses additional combinations, including an antibody comprising heavy-chain CDRs represented by SEQ ID NOs. 16–18 and light-chain CDRs represented by SEQ ID NOs. 25–27. The specification does not establish possession of this additional cross-paired combination as an antibody possessing the required APOA2 non-terminal binding specificity.

Regarding the broader antibody disclosures, the specification expressly recites alternative heavy-chain and light-chain CDR sequences corresponding to the claimed antibody categories ([0114]–[0117], p.). Nevertheless, the recitation of these alternatives does not, without sufficient supporting structural or functional disclosure, establish possession of every functional antibody encompassed by their independent combination. The specification identifies the constituent sequences and specifically disclosed antibody clones but does not identify representative cross-paired antibodies or establish a

structure–function relationship demonstrating that the additional combinations possess the required antigen-binding specificities.

Regarding the experimental evidence, the specification discloses that as a result of screening by ELISA in the same way as in the step (a), three hybridomas, clones 4C6-1, 5D9-3, and 6B4-2, specifically reacting only with the AT peptide, were obtained ([0124], p. 56). Paragraph [0124] further discloses that FIG. 1 shows the binding activity of clones 4C6-1, 5D9-3, and 6B4-2 against the APOA2-AT peptide and the APOA2-ATQ peptide. The specification further discloses that as a result of sequence analysis on genes encoding these monoclonal antibodies, the CDR sequences of 4C6-1 were shown to be the amino acid sequences represented by SEQ ID NOs: 33 to 38; the CDR sequences of 5D9-3 were shown to be the amino acid sequences represented by SEQ ID NOs: 39 to 44; and the CDR sequences of 6B4-2 were shown to be the amino acid sequences represented by SEQ ID NOs: 45 to 50 ([0125], p. 57).

Additionally, the specification discloses that FIG. 2 shows results of detecting each concentration of the APOA2-AT protein by sandwich ELISA using anti-APOA2 non-terminus monoclonal antibody MAB1 and each anti-APOA2-AT terminus monoclonal antibody or an anti-APOA2-AT terminus polyclonal antibody ([0128], p. 58). Paragraph [0128] further discloses that as shown in FIG. 2, increase in absorbance was confirmed in a manner dependent on the concentration of the APOA2-AT protein. Thus, Figures 1 and 2 demonstrate binding specificity and detection using the specifically disclosed antibodies. However, these figures do not establish the antigen-binding specificities of antibodies comprising heavy-chain and light-chain CDR triplets selected from different disclosed clones.

The specification further discloses that the CDRs are known to be conformationally complementary to the antigen molecule and to determine the specificity of the antibody ([0030], p. 17). Paragraph [0030] additionally discloses that in the immunoglobulin molecule, VL and VH are paired by dimerization to form an antigen-binding site. Accordingly, the specification recognizes that antigen-binding specificity depends on the structural relationship between antibody variable regions. However,

the specification does not establish that the additional heavy-chain and light-chain CDR combinations encompassed by claim 1 retain the required APOA2 binding specificities. The disclosure of individual CDR sequences, specific antibody clones, and independently recited CDR alternatives does not, without sufficient representative embodiments or structural information, reasonably convey possession of the full scope of functional antibodies encompassed by claim 1.

Claims 4, 6, and 8 depend from claim 1 and incorporate the same antibody limitations. The additional cutoff-value and discriminant limitations do not restrict the antibodies to the specifically disclosed heavy-chain and light-chain CDR pairings and therefore do not overcome the written-description deficiency identified for claim 1. Therefore, claims 1, 4, 6, and 8 fail to comply with the written description requirement of 35 U.S.C. §112(a).

> The following is a quotation of 35 U.S.C. 112(b):
> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

> The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claim **8** is rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Regarding claim 8, the claim depends from claim 1, which requires specific anti-APOA2-AT terminus, anti-APOA2-ATQ terminus, and anti-APOA2 non-terminus monoclonal antibodies defined by particular heavy-chain and light-chain CDR sequences. However, claim 8 recites an anti-APOA2-ATQ terminus antibody and an anti-APOA2-ATQ non-terminus antibody in step (A), and an anti-APOA2-AT terminus antibody and an anti-APOA2-AT non-terminus antibody in step (B), without specifying whether

these antibodies correspond to the CDR-defined antibodies required by claim 1 or constitute additional antibodies.

Accordingly, it is unclear whether steps (A) and (B) further define the antibody-mediated measurements recited in claim 1 or introduce separate antibody-mediated measurements using antibodies not subject to the specific CDR limitations recited in claim 1. This uncertainty concerning the relationship between the antibody limitations of the independent and dependent claims renders the scope of claim 8 unclear. For purposes of compact prosecution, claim 8 will be interpreted as requiring that the antibodies recited in steps (A) and (B) correspond to the respective anti-APOA2-ATQ terminus, anti-APOA2-AT terminus, and anti-APOA2 non-terminus antibodies recited in claim 1, including their applicable heavy-chain and light-chain CDR limitations. Appropriate correction is required.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

The rejection of claims 1, 6, and 8 under 35 U.S.C. §101 is maintained in view of the claim amendments. The rejection has been updated to address the amended claim limitations and to provide further analysis under Step 2A and Step 2B of the subject matter eligibility framework. The rejection of claim 4 under 35 U.S.C. §101 is withdrawn for the reasons discussed below.

Claims **1**, **6,** and **8** are rejected under 35 U.S.C. 101 because the claimed invention is directed to judicial exceptions, including a law of nature/natural phenomenon and an abstract idea, without reciting additional elements that are sufficient to amount to significantly more than the judicial exceptions.

This rejection is made in accordance with Patent Subject Matter Eligibility as set forth in MPEP §2106. Analysis of subject-matter eligibility under 35 U.S.C. §101 requires consideration under these steps as followed:

Step 1 – Statutory Category (Refer to MPEP §2106.03): Claim 1, 6, and 8 are drawn to a process, which falls within a statutory category under 35 U.S.C. §101.

Step 2A, Prong One – Recitation of a Judicial Exception (Refer to MPEP §2106.04): Regarding claim 1, the claim recites measuring the amount of APOA2-AT protein or APOA2-ATQ protein in a blood, plasma, or serum sample and determining that an intraductal papillary mucinous neoplasm (IPMN) is malignant when the APOA2-AT amount is lower than the amount associated with a known benign pancreatic cystic tumor or when the APOA2-ATQ amount is higher than the amount associated with a known benign pancreatic cystic tumor. The relationship between the naturally occurring amounts of APOA2 protein isoforms and the malignant or benign condition of an IPMN is a law of nature/natural phenomenon. The claimed determination relies upon the biological correlation between the measured protein amount and the pathological condition of the subject. Additionally, the claim recites an abstract idea in the form of a mental process involving observation, evaluation, and comparison of measured protein amounts with reference amounts to reach a diagnostic conclusion. The comparison and determination can be performed through mental evaluation of the measured values.

For purposes of the maintained rejection, claim 1 is evaluated based on the expressly recited alternative in which the amount of APOA2-ATQ protein is measured and malignant IPMN is determined when the measured APOA2-ATQ amount is higher than the corresponding amount associated with a known benign pancreatic cystic tumor. Because claim 1 recites the APOA2-AT and APOA2-ATQ diagnostic alternatives using “or,” the claim does not require both isoforms to be measured or both diagnostic relationships to be satisfied. The present rejection relies on the APOA2-ATQ alternative, rather than on a finding that the particular anti-APOA2-AT-terminal antibody configurations are conventional.

Regarding claim 6, the claim further recites determining malignancy when the APOA2-ATQ amount is higher than a cutoff value established using samples associated with known benign or malignant pancreatic cystic tumors. The relationship between the APOA2-ATQ amount and malignancy remains a natural correlation, while evaluating the measured amount against the cutoff constitutes an abstract idea in the form of a mental process.

Regarding claim 8, the claim further recites determining whether the subject has a pancreatic cystic tumor using an imaging technique and/or measuring APOA2-ATQ and APOA2-AT, inputting their measured amounts into a preset discriminant, and comparing the resulting discriminant value with that of a known normal subject. The claimed comparison and determination constitute an abstract idea in the form of a mental process involving mathematical evaluation and diagnostic interpretation of measured data. For purposes of the maintained rejection, claim 8 is evaluated based on the alternative requiring an imaging technique to determine in advance that the subject has a pancreatic cystic tumor, together with the APOA2-ATQ measurement and diagnostic determination alternative inherited from claim 1. Because claim 8 recites the imaging technique and the additional steps (A)–(C) using "and/or," the imaging alternative does not require performance of steps (A)–(C), including the additional APOA2-AT measurement recited therein. Accordingly, claims 1, 6, and 8 recite judicial exceptions.

Step 2A, Prong Two – Integration into a Practical Application (Refer to MPEP §2106.04 (d)): Regarding claim 1, the additional elements include obtaining a blood, plasma, or serum sample and measuring APOA2-AT or APOA2-ATQ using specified terminal and nonterminal antibodies or antigen-binding fragments. The claim specifies antibody CDR sequences for detecting the respective protein isoforms. However, the claim does not recite a particular improvement in the operation of an immunoassay, a modified assay mechanism, or a technological process that changes the biological relationship being evaluated. Rather, the antibodies are used to obtain the protein measurements subsequently evaluated according to the recited natural correlation. The antibody-mediated

measurements constitute data acquisition preceding the diagnostic determination. The claim does not require a particular treatment, alteration of the subject's condition, or other subsequent application of the diagnostic result. The determination is completed by comparing the measured amount with a reference amount and identifying the tumor as malignant.

Specifically, under the APOA2-ATQ alternative relied upon in this rejection, the recited anti-APOA2-ATQ-terminal and anti-APOA2 nonterminal antibodies are used to measure the amount of APOA2-ATQ protein, and the resulting measurement is evaluated according to the claimed relationship between an increased APOA2-ATQ amount and malignant IPMN. The antibody-mediated measurement supplies the information used in the diagnostic determination without requiring a further technological application of that determination. Accordingly, the additional elements do not integrate the judicial exceptions into a practical application.

Regarding claim 6, the additional cutoff-value limitations further specify how the measured APOA2-ATQ amount is evaluated. This limitation does not modify the antibody-mediated measurement process or require a technological application of the diagnostic result. Rather, it further defines the comparison used to evaluate the naturally occurring relationship between the protein amount and malignancy. Accordingly, claim 6 does not integrate the judicial exceptions into a practical application.

Regarding claim 8, the additional limitations include an imaging technique and/or further antibody-mediated measurements, inputting the measured values into a preset discriminant, and comparing the resulting value with that of a known normal subject. The imaging limitation obtains diagnostic information without requiring a particular imaging apparatus, imaging protocol, or technological improvement. The discriminant and comparison limitations process the measured values to reach a diagnostic conclusion. Although the claim recites particular antibody-based measurements, it does not require an improvement to the underlying imaging or measurement technology or a treatment or other application following the diagnostic determination. Under the imaging alternative relied upon in

this rejection, claim 8 requires determining in advance that the subject has a pancreatic cystic tumor using an imaging technique, without requiring the additional measurements and discriminant evaluation of steps (A)–(C). The imaging limitation supplies diagnostic information concerning the presence of a pancreatic cystic tumor, while the inherited APOA2-ATQ measurement and diagnostic determination limitations of claim 1 apply the recited natural correlation. The claim does not require the imaging technique to modify the antibody-mediated measurement process or produce a technological application of the resulting malignancy determination. Accordingly, claim 8 does not integrate the judicial exceptions into a practical application.

Step 2B, Inventive Concept (Refer to MPEP §2106.05): Regarding claim 1, the additional elements, considered individually and as an ordered combination, do not amount to significantly more than the identified judicial exceptions. As explained above, the present rejection is based on the APOA2-ATQ alternative expressly encompassed by claim 1, which requires measuring APOA2-ATQ using the recited anti-APOA2-ATQ-terminal and anti-APOA2 non-terminal antibodies and determining malignant IPMN based on the specified increase in APOA2-ATQ relative to the known benign pancreatic cystic tumor reference. The rejection does not depend upon establishing that the separately recited anti-APOA2-AT-terminal antibody configurations were well-understood, routine, and conventional.

First, regarding MPEP §2106.05(d), Honda demonstrates that antibody-mediated measurements of APOA2 isoforms in blood-derived samples were known in the field. Honda expressly discloses novel sandwich ELISAs for measuring these apoAII-isoforms using the newly established antibodies (p. 2) and the pan-apoAII goat polyclonal antibody was coated on the well surfaces of the microtiter plate as the capture antibody (p. 2, Fig. 1A). Honda further discloses the apoAII-ATQ–specific mouse monoclonal antibody was used for detection (p. 2, Fig. 1A), demonstrating antibody-mediated APOA2-ATQ measurement. Honda additionally describes the apoAII-AT–specific rabbit polyclonal antibody and the pan-apoAII mouse monoclonal antibody in the APOA2-AT assay (p. 2, Fig. 1B). Honda further establishes

the use of the assays with clinical samples, disclosing plasma and serum concentrations of apoAII-ATQ/AT were measured in three independent cohorts (p. 1) and the levels of apoAII-AT and -ATQ in the plasma of patients with IDACP were inversely correlated (p. 4, Fig. 2A).

Sanada provides more specific evidence concerning the antibody sequences recited in claim 1. Sanada expressly identifies anti-APOA2-ATQ-terminal monoclonal antibody CDR configurations comprising SEQ ID NOs: 4, 5, and 6 with SEQ ID NOs: 7, 8, and 9, or SEQ ID NOs: 10, 11, and 12 with SEQ ID NOs: 13, 14, and 15 ([0035]-[0036], p. 3). Sanada also identifies nonterminal monoclonal antibody CDR configurations comprising SEQ ID NOs: 16, 17, and 18 with SEQ ID NOs: 19, 20, and 21, or SEQ ID NOs: 22, 23, and 24 with SEQ ID NOs: 25, 26, and 27 ([0037]-[0038], p. 3). These CDR configurations correspond residue-for-residue to the respective APOA2-ATQ-terminal and nonterminal antibody configurations recited in claim 1. Because claim 1 permits measurement of APOA2-ATQ independently of APOA2-AT, the claim encompasses an alternative using antibody configurations expressly disclosed by Sanada.

Thus, the particular antibody configurations relied upon for the APOA2-ATQ alternative are not being treated as conventional merely because antibody-mediated detection techniques were generally known. Rather, Sanada provides evidence identifying the specific ATQ-terminal and nonterminal antibody configurations recited in the claim, while Honda provides evidence of antibody-mediated APOA2-ATQ measurement in clinical samples. These disclosures are relied upon in evaluating the claimed APOA2-ATQ measurement limitations under Step 2B.

Second, regarding MPEP §2106.05(f), the claimed measurements are used to obtain the APOA2 protein amounts necessary to apply the natural correlation. The claim does not require any further technological application of the resulting determination. The antibody sequences identify the reagents used to acquire the diagnostic data, but the claimed process ultimately uses those data to determine whether the IPMN is malignant. Third, regarding MPEP §2106.05(g), the sample acquisition and protein measurement steps serve as data-gathering activities preceding the diagnostic comparison. These steps

do not, in the claimed ATQ-only alternative, impose a meaningful limitation beyond obtaining the information required to evaluate the judicial exceptions.

When considered as an ordered combination, the additional elements provide antibody-mediated measurement followed by comparison of the measured protein amount with a reference amount to reach a diagnostic conclusion. The arrangement does not require an unconventional interaction among the additional elements that improves the assay technology or applies the diagnostic determination in a further technological process. In particular, the claimed APOA2-ATQ alternative employs the recited ATQ-terminal and nonterminal antibody configurations to obtain an APOA2-ATQ measurement and uses that measurement to evaluate the recited malignancy relationship. The present rejection is therefore directed to this specific alternative encompassed by claim 1, rather than to an assumption that all antibody configurations recited in claim 1 were conventionally used. Accordingly, claim 1 does not recite additional elements amounting to significantly more than the judicial exceptions.

Regarding claim 6, the additional elements, considered individually and in combination with claim 1, do not amount to significantly more than the identified judicial exceptions. The claim further requires determining malignancy when the APOA2-ATQ amount exceeds a cutoff established from samples associated with known benign or malignant pancreatic cystic tumors. Regarding MPEP §2106.05(d), Sanada expressly discloses the ATQ-terminal and nonterminal monoclonal antibody CDR configurations recited in claim 1, as discussed above. Honda establishes that APOA2 isoform measurements were evaluated against diagnostic reference values, disclosing that defined cut-off values for apoAII-ATQ/AT and CA19–9 (p. 6) and defined an apoAII-ATQ/AT level of less than 46.3 μg/ml and/or a CA19–9 level of more than 75 units/ml as IDACP (p. 7, Fig. 3I).

Regarding MPEP §§2106.05(f) and (g), the claimed cutoff further specifies the evaluation of the measured APOA2-ATQ amount. It does not require modification of the antibody-mediated measurement process or a subsequent technological application. The cutoff is used to determine whether the naturally

occurring biomarker amount indicates malignancy. Considered as an ordered combination, the additional elements provide an APOA2-ATQ measurement using specified antibodies followed by comparison with a diagnostic cutoff. The combination does not amount to significantly more than the judicial exceptions. Accordingly, claim 6 does not recite an inventive concept sufficient to confer patent eligibility.

Regarding claim 8, the additional elements, considered individually and in combination with claim 1, do not amount to significantly more than the identified judicial exceptions. The claim recites determining whether the subject has a pancreatic cystic tumor using an imaging technique and/or performing APOA2-ATQ and APOA2-AT measurements, inputting the measured amounts into a preset discriminant, and comparing the resulting discriminant value with that of a known normal subject. For purposes of the present rejection, claim 8 is evaluated based on the alternative requiring an imaging technique to determine in advance that the subject has a pancreatic cystic tumor, together with the APOA2-ATQ measurement and diagnostic determination alternative inherited from claim 1. Because claim 8 recites the imaging technique and the additional steps (A)–(C) using "and/or," the imaging alternative does not require performance of steps (A)–(C), including the additional APOA2-AT measurement recited therein. Accordingly, the present rejection does not rely on establishing that the particular anti-APOA2-AT-terminal antibody configurations were conventionally used.

Regarding MPEP §2106.05(d), Honda demonstrates that APOA2-ATQ and APOA2-AT measurements were used together to evaluate pancreatic disease. Honda discloses developing a formula to calculate the concentration of apoAII-ATQ/AT (p. 4) and a high correlation was observed between the results for apoAII-ATQ/AT obtained by MS and those calculated using equation-1 (p. 4, Fig. 1G). Honda further describes the AUC value based on the linear combination of log-transformed CA19–9 and apoAII-ATQ/AT, estimated from a linear logistic regression model (p. 8, Fig. 5E). These disclosures establish that mathematical evaluation of APOA2-related biomarker measurements for diagnostic classification was

known. They do not establish that the particular discriminant recited in claim 8 is identical to Honda's formula.

Regarding MPEP §2106.05(f), the imaging and antibody-mediated measurements obtain diagnostic information, while the discriminant and comparison limitations direct mathematical evaluation of that information. The claim does not require a particular improvement in imaging technology, an improved immunoassay mechanism, or a further technological application of the resulting diagnosis. Regarding MPEP §2106.05(g), the imaging and measurement activities supply the information used in the diagnostic evaluation. The discriminant and comparison steps do not alter the underlying biological sample or produce a technological result independent of the diagnostic determination. Considered as an ordered combination, the additional elements provide diagnostic data acquisition, mathematical processing of measured biomarker amounts, and comparison with a normal reference. The combination does not require a particular technological improvement or a further application of the diagnostic conclusion sufficient to transform the judicial exceptions into patent-eligible subject matter. Accordingly, claim 8 does not recite additional elements amounting to significantly more than the judicial exceptions.

Specifically, under the imaging alternative relied upon in the present rejection, the additional limitations provide imaging-based identification of a pancreatic cystic tumor before the APOA2-ATQ measurement and malignancy determination inherited from claim 1. The imaging technique supplies information concerning the presence of a pancreatic cystic tumor, while the antibody-mediated APOA2-ATQ measurement supplies the biomarker information used to evaluate the claimed natural correlation. Considered individually and as an ordered combination, these limitations do not require a particular improvement in imaging or immunoassay technology or a further technological application of the diagnostic determination.

Claims 1, 6, and 8, when considered individually and as a whole, are directed to judicial exceptions without reciting additional elements sufficient to amount to significantly more than those exceptions. Therefore, claims 1, 6, and 8 are rejected under 35 U.S.C. §101.

Closest Prior Art of Record

The closest prior art of record is Sanada et al. (US 20160245815 A1 – IDS dated 09/21/2023), which discloses a method for detecting a pancreatic tumor, i.e., pancreatic cancer or benign pancreatic tumor, comprising measuring the amounts of APOA2 protein variants in a sample of a test subject using antibodies specifically binding to APOA2 proteins. Sanada discloses separately measuring APOA2-ATQ and APOA2-AT protein variants using sandwich ELISA comprising antibodies specifically binding to the respective C-terminal regions and antibodies binding to regions other than the C-terminal regions. Sanada expressly teaches anti-APOA2-AT terminus antibodies, including monoclonal antibodies, and anti-APOA2 protein non-terminus antibodies. Furthermore, Sanada discloses anti-APOA2-ATQ terminus monoclonal antibodies having CDR sequences represented by SEQ ID NOs: 4–9 or 10–15, and anti-APOA2 protein non-terminus monoclonal antibodies having CDR sequences represented by SEQ ID NOs: 16–21 or 22–27. These correspond to the APOA2-ATQ terminus and APOA2 non-terminus antibody CDR limitations recited in amended claim 1. However, Sanada does not disclose the particular anti-APOA2-AT terminus monoclonal antibody CDR sequences represented by SEQ ID NOs: 33–38, 39–44, or 45–50, as recited in amended claim 1.

Additionally, although Sanada discloses detecting pancreatic cancer and benign pancreatic tumors and distinguishing these conditions using APOA2 measurements in combination with additional tumor markers (p. 22, Examples 10–11, [0236]–[0240]), Sanada does not expressly teach determining malignant intraductal papillary mucinous neoplasm (IPMN) based on an APOA2-AT amount lower than, or an APOA2-ATQ amount higher than, the corresponding amount obtained from a subject having a known

benign pancreatic cystic tumor. Accordingly, Sanada represents particularly close prior art because it discloses the relevant APOA2 biomarkers, antibody-based detection methods, APOA2-AT-specific antibodies, and specifically recited APOA2-ATQ and non-terminus antibody CDR sequences, but does not expressly disclose the claimed malignant-IPMN determination or the particular APOA2-AT antibody CDR sequences.

Response to Arguments

Applicant's arguments filed, 08/21/2026, have been fully considered. The response is set forth below.

Rejection Under 35 U.S.C. §112(a) — Enablement

Applicant's arguments concerning the previous rejection of claims 1–8 under 35 U.S.C. §112(a), based on the enablement requirement, have been considered but are moot because the previous rejection has been withdrawn in view of the amendments.

However, upon further consideration of the amended claims, a new rejection under 35 U.S.C. §112(a), based on the written-description requirement, is made in the present Office Action. As set forth in the present Office Action, although the specification discloses individual antibody CDR sequences and particular antibody configurations, the specification does not reasonably convey possession of the full scope of heavy-chain and light-chain CDR combinations encompassed by amended claim 1. Specifically, the specification identifies particular heavy-chain and light-chain CDR groupings associated with individual anti-APOA2-AT terminus, anti-APOA2-ATQ terminus, and anti-APOA2 non-terminus monoclonal antibodies. However, the amended claim language encompasses additional combinations of the independently recited heavy-chain and light-chain CDR alternatives.

The specification does not provide sufficient representative species, structural guidance, or characterization demonstrating possession of these additional combinations while retaining the claimed

APOA2-binding specificity. Accordingly, the rejection under 35 U.S.C. §112(a), based on the written-description requirement, is maintained as updated in the present Office Action.

Rejection Under 35 U.S.C. §101

Applicant's arguments concerning the previous rejection of claims 1-8 under 35 U.S.C. 101 has been fully considered but they are not persuasive with respect to claims 1, 6, and 8. Applicant argues that amended claim 1 satisfies Step 2A, Prong Two and Step 2B because the additional limitations integrate the recited natural relationship into a practical application and provide significantly more than the judicial exception. Applicant particularly contends that the claimed method addresses the invasiveness and difficulty associated with conventional pancreatic cystic tumor diagnosis by measuring APOA2-AT or APOA2-ATQ in blood, plasma, or serum using antibodies having particular CDR sequences. These arguments are not persuasive with respect to the APOA2-ATQ measurement alternative relied upon in the maintained rejection, for the following reasons.

Regarding Step 2A, Prong Two, the Examiner acknowledges that amended claim 1 recites determining malignant intraductal papillary mucinous neoplasm (IPMN), obtaining a blood, plasma, or serum sample, measuring APOA2-AT or APOA2-ATQ protein amounts using specified antibodies, and comparing the measured amounts with corresponding amounts obtained from a subject having a known benign pancreatic cystic tumor. However, these additional limitations do not integrate the judicial exception into a practical application. The claimed diagnostic determination is based on the naturally occurring relationship between APOA2 protein concentrations and malignant pancreatic cystic tumors. In particular, claim 1 recites determining malignancy when the measured APOA2-AT amount is lower than the corresponding amount in a subject having a known benign pancreatic cystic tumor or when the measured APOA2-ATQ amount is higher than the corresponding amount in such a subject.

For purposes of the maintained rejection, the Examiner relies on the APOA2-ATQ measurement alternative, wherein malignant IPMN is determined when the measured APOA2-ATQ amount is higher

than the corresponding amount associated with a known benign pancreatic cystic tumor. Because amended claim 1 recites the APOA2-AT and APOA2-ATQ alternatives using "or," the claim does not require measurement of both isoforms or satisfaction of both diagnostic relationships.

The recited sample collection, antibody-based measurement, and comparison steps provide the information used to evaluate this naturally occurring relationship. The claim does not require a particular improvement in the operation of an immunoassay instrument, a new measurement mechanism, or a technological transformation resulting from the diagnostic determination. Although Applicant argues that the method provides a convenient and noninvasive diagnostic approach, the asserted clinical benefit concerns obtaining and interpreting diagnostic information rather than an improvement to the underlying measurement technology. Accordingly, the additional limitations do not integrate the judicial exception into a practical application under Step 2A, Prong Two.

Regarding Step 2B, Applicant argues that the recitation of particular anti-APOA2 antibodies having specified CDR sequences provides significantly more than the judicial exception. The Examiner has considered these limitations individually and as an ordered combination, including the alternative antibody configurations encompassed by amended claim 1. Importantly, amended claim 1 recites measuring APOA2-AT or APOA2-ATQ. Thus, the claim encompasses alternative diagnostic embodiments involving APOA2-AT measurement alone or APOA2-ATQ measurement alone. Each alternative must be considered in determining whether the claim as a whole recites additional elements sufficient to provide significantly more than the judicial exception. The maintained rejection is based on the APOA2-ATQ measurement alternative, which encompasses the antibody configurations expressly disclosed by Sanada. The rejection does not depend upon establishing that the separately recited anti-APOA2-AT-terminal antibody configurations were well-understood, routine, and conventional.

Regarding the APOA2-ATQ measurement alternative, amended claim 1 permits determining malignant IPMN based on an APOA2-ATQ amount higher than the corresponding amount obtained from

a subject having a known benign pancreatic cystic tumor. This alternative does not require measuring APOA2-AT or employing the specifically recited anti-APOA2-AT terminus monoclonal antibody CDR sequences represented by SEQ ID NOs: 33–50. Rather, the APOA2-ATQ alternative employs an anti-APOA2-ATQ terminus antibody and an anti-APOA2 non-terminus antibody, or their binding fragments, to measure APOA2-ATQ. As discussed in the present Office Action, Sanada discloses antibody-based measurement of APOA2-ATQ using anti-APOA2-ATQ terminus and anti-APOA2 non-terminus antibodies, including the corresponding monoclonal antibody CDR configurations recited in amended claim 1. Accordingly, the recitation of the particular anti-APOA2-AT terminus antibody CDR sequences does not establish significantly more for the independently encompassed APOA2-ATQ measurement alternative because those antibody sequences are not required to practice that alternative.

Specifically, Sanada discloses anti-APOA2-ATQ-terminal monoclonal antibody CDR configurations comprising SEQ ID NOs: 4–9 or 10–15 and anti-APOA2 nonterminal monoclonal antibody CDR configurations comprising SEQ ID NOs: 16–21 or 22–27 ([0035]–[0038]). These configurations correspond to the respective antibody configurations recited in amended claim 1. Honda further establishes that antibody-mediated measurement of APOA2-ATQ in blood-derived samples was known. Thus, the Examiner relies on the specific antibody configurations disclosed by Sanada and the antibody-mediated measurement techniques disclosed by Honda in evaluating the APOA2-ATQ alternative under Step 2B, rather than treating all antibodies recited in claim 1 as conventional.

Furthermore, the recited APOA2-ATQ antibody-based measurement techniques perform their established function of selectively binding and measuring the corresponding APOA2 protein variant. The measured amount is subsequently compared with a reference amount to evaluate the naturally occurring relationship between APOA2-ATQ concentration and malignancy. The claim does not require the recited antibodies to perform a new technological function or produce a technical improvement beyond their use as selective immunoassay reagents.

Regarding the APOA2-AT measurement alternative, amended claim 1 permits determining malignant IPMN based on an APOA2-AT amount lower than the corresponding amount obtained from a subject having a known benign pancreatic cystic tumor. The Examiner acknowledges that this alternative recites anti-APOA2-AT terminus monoclonal antibody CDR sequences represented by SEQ ID NOs: 33–50. However, unlike the APOA2-ATQ alternative, the present evidence does not establish that the complete anti-APOA2-AT-terminal six-CDR antibody configurations represented by SEQ ID NOs: 33–38, 39–44, or 45–50 were well-understood, routine, and conventionally used for APOA2-AT detection. Although Honda establishes that APOA2-AT immunoassays were known, Honda does not disclose the particular claimed monoclonal antibody configurations. Accordingly, the Examiner does not rely on the conventionality of these particular configurations to maintain the §101 rejection of claim 1. Rather, the maintained rejection relies on the independently encompassed APOA2-ATQ alternative, as discussed above.

Regarding claim 4, the rejection under 35 U.S.C. §101 is withdrawn. Claim 4 requires the APOA2-AT measurement pathway and therefore requires consideration of the particular anti-APOA2-AT-terminal antibody configurations recited in claim 1. Because the present evidence does not establish that these complete six-CDR configurations were conventionally used for APOA2-AT detection, the record does not sufficiently support maintaining the §101 rejection of claim 4 under Step 2B. This withdrawal does not affect the separate rejection of claim 4 under 35 U.S.C. §112(a).

Regarding claim 6, the claim requires the APOA2-ATQ measurement pathway and further specifies a diagnostic cutoff. As discussed in the updated rejection, Sanada discloses the corresponding APOA2-ATQ-terminal and nonterminal antibody CDR configurations, while Honda establishes that APOA2 isoform measurements were evaluated against diagnostic reference values. The additional cutoff limitation specifies how the measured APOA2-ATQ amount is evaluated to determine malignancy. Considered individually and as an ordered combination with the inherited limitations of claim 1, the

additional elements do not amount to significantly more than the judicial exceptions. Accordingly, the §101 rejection of claim 6 is maintained.

Regarding claim 8, the rejection under 35 U.S.C. §101 is maintained based on the alternative requiring an imaging technique to determine in advance that the subject has a pancreatic cystic tumor, together with the APOA2-ATQ measurement and diagnostic determination alternative inherited from claim 1. Claim 8 recites the imaging technique and the additional steps (A)–(C) using "and/or." Accordingly, the imaging alternative does not require performance of steps (A)–(C), including the additional APOA2-AT measurement recited therein. The imaging technique obtains diagnostic information concerning the presence of a pancreatic cystic tumor, while the inherited APOA2-ATQ measurement and diagnostic determination limitations apply the recited natural correlation. The imaging limitation does not require a particular improvement in imaging technology or a further technological application of the resulting diagnosis. Considered individually and as an ordered combination, these limitations do not amount to significantly more than the judicial exceptions.

Accordingly, the ordered combination of obtaining a biological sample, measuring APOA2-ATQ using the specified antibodies, comparing the resulting amount with a reference amount, and determining malignant IPMN does not provide significantly more than the judicial exception for the APOA2-ATQ alternative relied upon in the maintained rejection of claim 1.

Applicant further contends that the particular antibody configurations effect a transformation or reduction of an article to a different state or thing. However, the recited antibodies are used to selectively bind and measure APOA2 protein variants. The claim does not require a particular transformation of the biological sample into a different article or state that constitutes a technological application of the judicial exception. The binding interactions associated with antibody-based measurement do not, merely by occurring during an immunoassay, establish that the claimed diagnostic determination integrates the judicial exception into a practical application or provides significantly more. Therefore, Applicant's

arguments concerning the particular antibody CDR sequences, noninvasive diagnostic advantages, and asserted practical application do not overcome the rejection of claims 1, 6, and 8. The additional elements, considered individually and as an ordered combination, do not amount to significantly more than the judicial exception for the reasons set forth in the present Office Action. Accordingly, the rejection of claims 1, 6, and 8 under 35 U.S.C. §101 is maintained as updated, while the rejection of claim 4 under 35 U.S.C. §101 is withdrawn.

Rejection Under 35 U.S.C. §103

Applicant's arguments concerning the previous rejection of claims 1–8 under 35 U.S.C. §103 over Honda in view of Okusaka, UniProt, Talar, Maire, Sanada, and Van Regenmortel have been considered but are moot because the previous rejection has been withdrawn in view of the amendments.

Nonstatutory Double Patenting

Applicant's arguments concerning the previous nonstatutory double-patenting rejection of claims 1–8 over claims 1–6 of U.S. Patent No. 11,162,956, in view of Honda, Okusaka, UniProt, Talar, Maire, Sanada, and Van Regenmortel, have been considered but are moot because the previous rejection has been withdrawn in view of the amendments.

Ending

Applicant's arguments and amendments have been fully considered. The previous rejection under 35 U.S.C. §112(a), based on the enablement requirement, is withdrawn. A new rejection under 35 U.S.C. §112(a), based on the written-description requirement, is maintained against claims 1, 4, 6, and 8. The rejection under 35 U.S.C. §101 is maintained against claims 1, 6, and 8 and withdrawn with respect to claim 4. The previous rejections under 35 U.S.C. §103 and nonstatutory double patenting are withdrawn.

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ELIZABETH OGUNTADE whose telephone number is (571)272-6802. The examiner can normally be reached Monday-Friday 6:00 AM - 3 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Bao-Thuy Nguyen can be reached at 571-272-0824. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/E.O./
Examiner, Art Unit 1677

/BAO-THUY L NGUYEN/
Supervisory Patent Examiner, Art Unit 1677
October 27, 2026